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Information entropy-based classification of triterpenoids and steroids from *Ganoderma*

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ABSTRACT

A set of 71 triterpenoid and steroid compounds from *Ganoderma* were periodically classified using a procedure based on information entropy with artificial intelligence. Six features were used in hierarchical order to classify the triterpenoids and steroids structurally. The phytochemicals belonging to the same group in the periodic table present similar antioxidant activity, and those compounds belonging to the same period exhibit maximum resemblance. The periodic classification is related to the experimental bioactivity and antioxidant potency data that are available in the literature: a steroid with a three-ketone group conjugated with two carbon–carbon double bonds in the right side of the periodic table exhibits the greatest antioxidant activity.

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1. Introduction

Ganoderma resembles a therapeutic fungal biofactory. This fungus induces apoptosis and inhibits cell proliferation. The comprehensive characterization of the compounds in *Ganoderma* allowed the identification of its major pharmacologically active constituents: (1) triterpenoids and steroids, (2) polysaccharides, (3) proteins, peptides and amino acids, (4) nucleosides, nucleotides and ribonucleic acids, and (5) alkaloids, vitamins, essential minerals, flavors and fatty acids (Paterson, 2006). *Ganoderma lucidum* and its active compounds were reviewed (Boh et al., 2007; Chinsembu and Hedimbi, 2010; Liang et al., 2014; Ma et al., 2011).

The structure classification system used here is not a brand-new finding and has been widely applied for the analysis of other types of phytochemicals. In earlier publications, the molecular classification of 33 phenolic compounds derived from the cinnamic and benzoic acids based on chemical structure indices and their relationship to the antioxidant properties of *Posidonia oceanica* (Castellano et al., 2012), the categorization of 74 flavonoid molecules using information theory (Castellano et al., 2013) and the clustering of 66 stilbenoids by information entropy (Castellano et al., 2014) were reported and related to their antioxidant activity.

In the present report, 71 triterpenoids and steroids from *Ganoderma* were classified using this information entropy-based algorithm. The aim of this study was to predict the antioxidant potency of any such substances from *Ganoderma* based on their molecular structures. A practical classification would provide a clearer understanding of these compounds.

The anti-inflammatory triterpenoids and steroids from *G. lucidum* and *Ganoderma tsugae* were reported, and the results show that a 3-ketone (C=O) compound presents potent, concentration-dependent, inhibitory activity on the release of β-glucuronidase stimulated with formyl-Met-Leu-Phe/cytochalasin B (fMLP/CB). The oxidation of the alcohol C-3α-OH moiety formed from the saponification of the acetic acid ester C-3-OAc enhanced the inhibitory potency, and the C-3-OAc compound exhibited significant inhibitory activity on the generation of superoxide anion $O_2^{\cdot-}$ induced by fMLP/CB (Ko et al., 2008). Moreover, the 24,25-dihydroxy and 5-OH substitutions decreased the inhibition of the generation of $O_2^{\cdot-}$. A similar tendency was found for the inhibition of the release of elastase (Thang et al., 2013). Some compounds (ganodermanondiol, lucidumol B, ganodermanontriol, ganoderic acid β and ganolucidic acid A in Table 1) showed significant anti-human immunodeficiency virus type-1 (HIV-1) protease activity. The hydroxyl (OH) groups at C-3 or at both C-24 and C-25 of a lanostane-type triterpene are essential moieties, showing the potency of the compounds against the HIV-1 protease (Min et al., 1998).

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Table 1
Vector property for triterpenoids and steroids from *Ganoderma*.

1	Δ^1-Lupenone	011111
2	Ergosta-7,22-dien-3β-ol	001011
3	Ergosta-4,6,8(14),22-tetraen-3-one	111111
4	Ergosterol peroxide	001011
5	Ganodermanondiol	000111
6	Lucidumol B	000011
7	Ergosta-7,22-dien-3-one	001111
8	3β,5α-Dihydroxy-(22*E*,24*R*)-ergosta-7,22-dien-6-one	001011
9	Ergosta-7,22-dien-3β-yl palmitate	001001
10	Ergosta-7,22-dien-2β,3α,9α-triol	001011
11	Ganoderal A	001110
12	Ganoderal B	001011
13	Ganoderic aldehyde A	001010
14	Tsugaric acid A	001001
15	3-Oxo-5α-lanosta-8,24-dien-21-oic acid	001111
16	3α-Acetoxy-5α-lanosta-8,24-dien-21-oic acid ester β-D-glucoside	001001
17	Tsugaric acid B	001001
18	Ganopsoreric acid A	001110
19	Ganoderic acid B	001010
20	Ganoderic acid C_1	001110
21	Ganoderic acid E	001110
22	Ganodermanontriol	000111
23	Lucidumol A	000111
24	Ganoderic acid β	001010
25	Ganoderiol F	001111
26	Ganoderic acid A	001110
27	Ganolucidic acid A	001110
28	Ganoderic acid γ	001110
29	Ganoderic acid δ	001110
30	Ganoderic acid ε	001010
31	Ganoderic acid ξ	001010
32	Ganoderic acid η	001010
33	Ganoderic acid θ	001010
34	Ganolucidic acid D	001110
35	Ganoderic acid C_2	001010
36	Lucidenic acid SP1	001010
37	Ganoderic acid C_6	001010
38	Ganoderic acid G	001010
39	Ganoderic acid D	001110
40	Ganoderic acid H	001010
41	Methyl ganoderate A	001111
42	Methyl ganoderate B	001011
43	11α-Hydroxy-3,7-dioxo-5α-lanosta-8,24(*E*)-dien-26-oic acid	001110
44	11β-Hydroxy-3,7-dioxo-5α-lanosta-8,24(*E*)-dien-26-oic acid	001110
45	12β-Acetoxy-7β-hydroxy-3,11,15,23-tetraoxo-5α-lanosta-8,20-dien-26-oic acid	001110
46	4,4,14α-Trimethyl-3,7-dioxo-5α-chol-8-en-24-oic acid	001110
47	12β-Acetoxy-3,7,11,15,23-pentaoxo-5α-lanosta-8-en-26-oic acid ethyl ester	001111
48	3β,7β-Dihydroxy-12β-acetoxy-11,15,23-trioxo-5α-lanosta-8-en-26-oic acid methyl ester	001011
49	Ganodermadiol	001011
50	Ganoderic acid DM	001110
51	Ganoderenic acid F	001110
52	Lucidadiol	001011
53	15α-Hydroxy-3-oxo-5α-lanosta-7,9,24(*E*)-trien-26-oic acid	001110
54	15α,26-Dihydroxy-5α-lanosta-7,9,24(*E*)-trien-3-one	001111
55	3β-Hydroxy-5α-lanosta-7,9,24(*E*)-trien-26-oic acid	001010
56	3β-Hydroxy-7-oxo-5α-lanosta-8,24(*E*)-dien-26-oic acid	001010
57	Lucideric acid A	001110
58	Lucidone A	001011
59	Ganolucidic acid E	001110
60	Ganoderic acid F	001110
61	Ganoderenic acid D	001110
62	Ganoderic acid J	001110
63	7β,12β-Dihydroxy-3,11,15,23-tetraoxo-5α-lanosta-8-en-26-oic acid	001110
64	12β-Hydroxy-3,7,11,15,23-pentaoxo-5α-lanosta-8-en-26-oic acid	001110
65	Ganoderenic acid B	001010
66	12β-Acetoxy-3β,7β-dihydroxy-11,15,23-trioxo-5α-lanosta-8,20-dien-26-oic acid	001110
67	Methyl lucidenate C	001011
68	Ganoderic acid AM	001010
69	Ganoderic acid T-Q	001110
70	Ganoderic acid K	001010
71	Ganoderenic acid A	001110

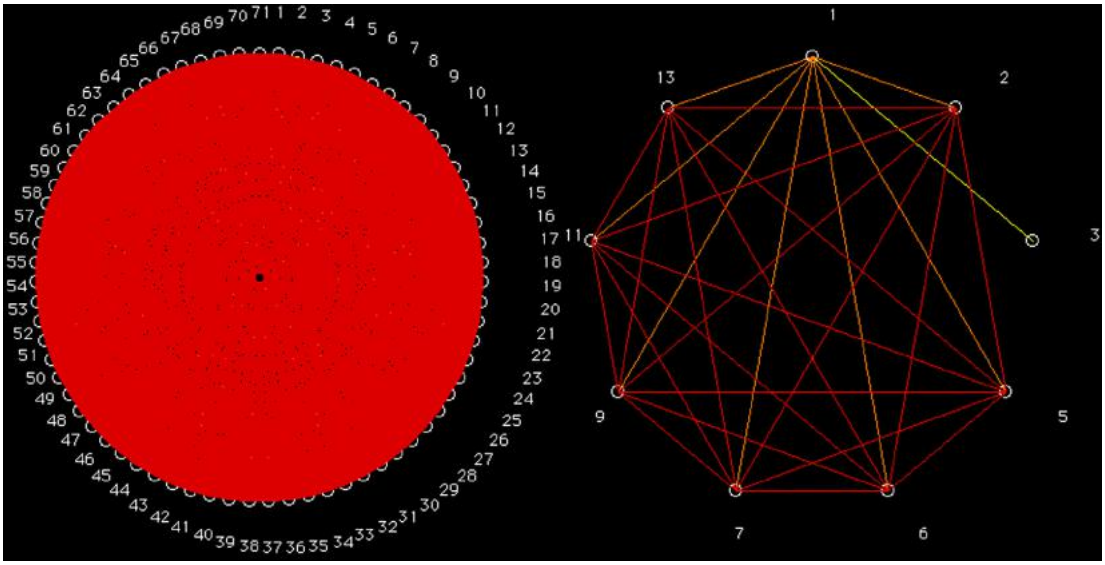


Fig. 1. Partial correlation diagram and fragment: high (*red*); medium (*orange*); low (*yellow*) correlations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The anti-HIV and anti-hepatitis C virus (HCV) compounds from natural sources were reviewed (Hattori et al., 2013).

The identified moieties of triterpenoids from *G. lucidum* result in increased activity: ganoderic alcohols (opposite to ganoderic acids), an OH substituted in position 26, an esterified form of carboxylic acid (COOH) group in the side chain, a C=O in position 3, and two conjugated C=C double bonds placed at positions 7 and 9. However, other moieties decrease the cytotoxic potency: the increase in the number of OHs in the side chain and in the ring skeleton and the acetylation of the 15-OH (Cheng et al., 2010).

Based on the above-described reasons, *i.e.*, the primary structural features of triterpenes and steroids, attention should be paid to certain structural requirements for efficient antioxidant activity to propose the classification.

2. Computational method

The computational method was previously described and successfully applied to the classifications of polyphenolic (Castellano et al., 2012), flavonoid (Castellano et al., 2013) and stilbenoid (Castellano et al., 2014) compounds. The first step for quantifying the concept of similarity for triterpenoid and steroid molecules is to list their most important moieties with respect to the antioxidant activity of the substances. Furthermore, the vector of properties $i = < i_1,i_2,\ldots i_k,\ldots >$ should be associated with every compound i, the components of which correspond to a number of characteristic groups in the molecule presented in hierarchical order according to the expected importance of their antioxidant potency. The components i_k are either "1" or "0" in concordance with the experimental conclusions of the antioxidant activity to structural variations in the substance: index $i_1 = 1$ indicates the presence of a C=O at position 3 conjugated with at least two C=C ($i_1 = 0$, otherwise), $i_2 = 1$ denotes the presence of a C=O at position 3 conjugated with at least one C=C, $i_3 = 1$ marks the absence of OH at both positions 24 and 25, $i_4 = 1$ connotes the presence of a C=O at position 3, $i_5 = 1$ shows the presence of either an OH or C=O at position 3, and $i_6 = 1$ indicates the absence of both an aldehyde (CHO) group and a COOH at positions 23 and 25. Ergosta-4,6,8(14),22-tetraen-3-one exhibits a C=O at position 3 conjugated with two C=C, the absence of OH at both positions 24 and 25, and the lack of both a CHO and a COOH at positions 23 and 25; thus, its associated vector is <1 1 1 1 1 1>. In this study, ergosta-4,6,8(14),22-tetraen-3-one was selected as a *reference* because of its markedly high antioxidant potency. Table 1 shows the vectors associated with 71 triterpenoids and steroids from *Ganoderma*. The <0 1 1 1 1 1> vector is associated with Δ^1-lupenone because it presents a C=O at position 3 conjugated with one C=C, the absence of OH at both positions 24 and 25, and a lack of both a CHO and a COOH at positions 23 and 25.

The similarity matrix $\mathbf{R} = [r_{ij}]$ between two triterpenoids or steroids, $i = < i_1,i_2,i_3,i_4,i_5,i_6>$ and $j = <j_1,j_2,j_3,j_4,j_5,j_6>$, is defined by the correlation coefficient between these compounds modified by six weights, such as 0.5, 0.25, 0.125. For instance, for molecule 1 (Δ^1-lupenone) <011111>, $r_{11} = 1{\cdot}1/2 + 1{\cdot}1/4 + 1{\cdot}1/8 + 1{\cdot}1/16 + 1{\cdot}1/32 + 1{\cdot}1/64 = 0.984375$ (identical to itself in all six indices), and for compounds 1 and 2 (ergosta-7,22-dien-3β-ol) <001011>, $r_{12} = 1{\cdot}1/2 + 0{\cdot}1/4 + 1{\cdot}1/8 + 0{\cdot}1/16 + 1{\cdot}1/32 + 1{\cdot}1/64 = 0.671875$ (concurrent only in indices 1, 3, 5 and 6).

Some learning procedures similar to those encountered in the stochastic methods are implemented as follows (White, 1989). Consider the given classification "*good*" obtained from practical observations, which corresponds to a reference similarity matrix $\mathbf{S} = [s_{ij}]$ obtained for an arbitrary number of properties. Matrix **S** is calculated in the same way as matrix **R**. Next, consider the same set of species as in the good classification and the actual properties.

The number of properties for two similarity matrices **R** and **S** may differ. The learning procedure consists of finding the classification results for **R** that are as close as possible to the good grouping. The distance between the partitions into the classes characterized by **R** and **S** is given by the following equation:

$$D = -\sum_{ij}(1 - r_{ij})\ln\frac{1 - r_{ij}}{1 - s_{ij}} - \sum_{ij} r_{ij}\ln\frac{r_{ij}}{s_{ij}} \quad \forall 0 \leqslant r_{ij}, s_{ij} \leqslant 1 \tag{1}$$

which is the definition suggested by the information theory introduced by Kullback (1959) to measure the distance between two probability distributions. The matrix of intra (r_{ii}) and intercorrelations ($r_{i\neq j}$) is the similarity matrix $\mathbf{R}$ $[r_{ij}]$. The procedure was applied to the synthesis of complex dendrograms using the information entropy (Iordache, 2011, 2012, 2014).

We wrote the MolClas program for molecular classification based on the equipartition conjecture of entropy production. This

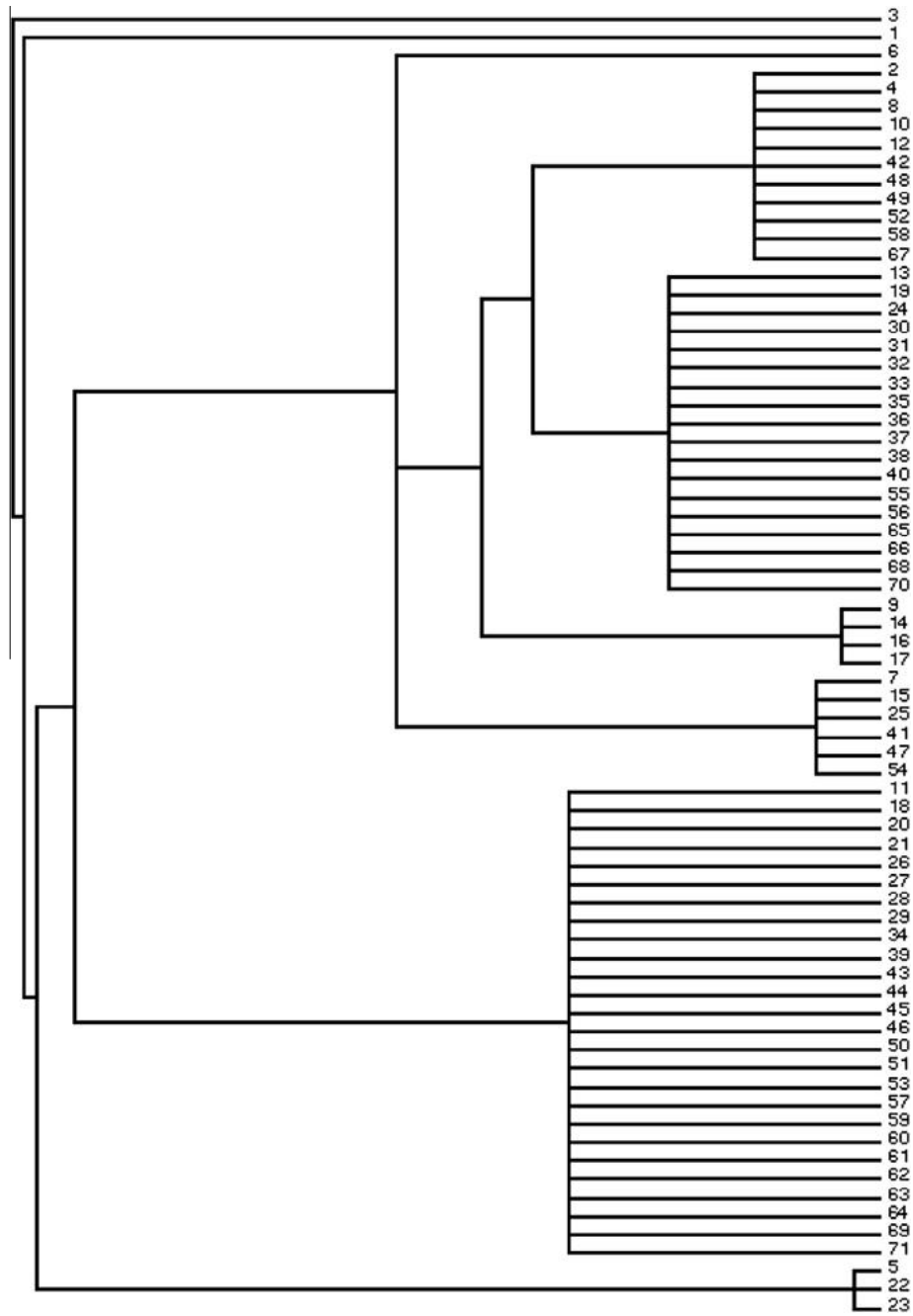


Fig. 2. Dendrogram of the studied triterpenoids and steroids.

Table 2
Classification level, number of classes and entropy for triterpenoids and steroids from *Ganoderma*.

Classification level *b*	Number of classes	Entropy *h*
1.00	71	1277.81
0.98	9	32.06
0.96	7	20.10
0.95	6	14.82
0.92	5	10.26
0.90	4	6.79
0.79	3	3.87
0.54	2	1.55
0.48	1	0.08

program produces the similarity and difference matrices in the NEXUS format (.nex) for the PAUP, MacClade and Splits-Tree algorithms. The MolClas procedure performs the single- and complete-linkage hierarchical cluster analyses (CAs) of the compounds using the IMSL subroutine CLINK (IMSL, 1989).

Our program GraphCor draws the partial correlation diagram, which presents high partial correlations ($r \geqslant 0.75$) in red, medium partial correlations ($0.50 \leqslant r < 0.75$) in orange, low partial correlations ($0.25 \leqslant r < 0.50$) in yellow and *zero* partial correlations ($r < 0.25$) in black. Both algorithms, MolClas and GraphCor, are available from the authors at Internet (torrens@uv.es) and are free for academics.

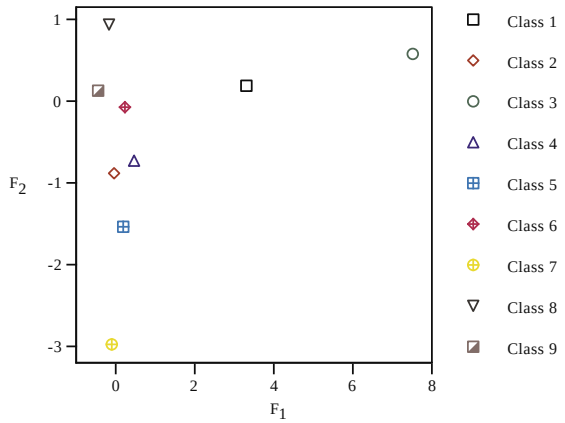


Fig. 3. Principal component analysis F_2 *vs.* F_1 scores plot for the studied triterpenoids and steroids.

3. Calculation results and discussion

3.1. Partial correlation diagram, dendrogram, principal component analysis and the periodic table

A dataset of 71 triterpenoids and steroids from *Ganoderma* (Table 1) were obtained from the technical literature (Cheng et al., 2010; Ko et al., 2008; Ma et al., 2011; Thang et al., 2013). The matrix of the Pearson correlation coefficients between every pair of vector properties $<i_1,i_2,i_3,i_4,i_5,i_6>$ of the 71 triterpenoid and steroid derivatives was calculated. The pairs of compounds with high partial correlations show a similar vector property. After the equipartition conjecture, the partial correlation matrix (*cf.* Fig. 1) contains 2346 high, 69 medium, one low and 69 *zero* partial correlations. The red lines link the phytochemicals with the greatest activity in the same class and thus represent high partial correlations. A subset of nine substances {1–3,5–7,9,11,13} with relatively great antioxidant activity was selected, and these correspond to 21 high, seven medium, one low and seven *zero* partial correlations. For instance, molecule 2 (ergosta-7,22-dien-3 β-ol) shows high partial correlations with compounds 5–7, 9, 11 and 13 ($r \geqslant 0.75$, Fig. 1b, *red*) and medium partial correlation with substance 1 ($0.50 \leqslant r < 0.75$, *orange*).

The grouping rule in the case with equal weights $a_k = 0.5$ for the classification level $0.97 \leqslant b \leqslant 0.98$ allows the following clusters: C_{b9} = (1)(2,4,8,10,12,42,48,49,52,58,67)(3)(5,22,23)(6)(7,15,25,41,47,54)(9,14,16,17)(11,18,20,21,26–29,34,39,43–46,50,51,53,57,59–64,69,71)(13,19,24,30–33,35–38,40,55,56,65,66,68,70)

Nine classes were obtained with the associated entropy $h(\mathbf{R}_{b9}) = 32.06$. The dendrogram (Jarvis and Patrick, 1973; Page, 2000) matching $<i_1,i_2,i_3,i_4,i_5,i_6>$ and C_{b9} (*cf.* Fig. 2) was calculated (Shaw, 2003). In particular, the triterpenoids and steroids with the greatest activity (1,3) were grouped into close clusters. The dendrogram separates the same nine classes, in agreement with the partial correlation diagram (Fig. 1) and C_{b9}.

At level $b = 0.96$, the set of groupings was found to be the following: C_{b7} = (1)(2,4,8,10,12,13,19,24,30–33,35–38,40,42,48,49,52,55,56,58,65–68,70)(3)(5,22,23)(6)(7,11,15,18,20,21,25–29,34,39,41,43–47,50,51,53,54,57,59–64,69,71)(9,14,16,17)

Seven classes were found as a result, and the entropy decays to $h(\mathbf{R}_{b7}) = 20.10$. Again, the triterpenoids and steroids with the greatest activity (1,3) were grouped into close clusters. The dendrogram separates seven classes, in qualitative agreement with the partial correlation diagram (Fig. 1), C_{b7} and C_{b9}.

A comparative analysis of the set from 71 individual classes (every compound in its own grouping) to one cluster (containing all substances) is summarized in Table 2, in agreement with the partial correlation diagram (Fig. 1), C_{b7} and C_{b9}.

In the principal component analysis (PCA) F_2–F_1 scores plot, the triterpenoids and steroids with the same vector property collapse. The same nine classes of triterpenoids and steroids were clearly distinguished in concordance with C_{b9}: class 1 ($F_1 > F_2 \approx 0$, *cf.* Fig. 3, *top*), grouping 2 ($0 \approx F_1 > F_2$, *left*), cluster 3 ($F_1 > F_2 > 0$, *right*), class 4 ($F_1 >> F_2$, *left*), grouping 5 ($F_1 >> F_2$, *left*), cluster 6 ($F_1 >> F_2$, *left*), class 7 ($0 > F_1 > F_2$, *bottom*), grouping 8 ($F_1 << F_2$, *top left*) and cluster 9 ($F_1 << F_2$, *left*). This classification is in qualitative agreement with the partial correlation diagram and dendrogram (Figs. 1 and 2).

Table 3
Table of periodic pharmacological activities of triterpenoids and steroids from *Ganoderma*. Data are IC_{50} (μM) or percentage of inhibition ± standard error of the mean.

	g0000	g0001	g0010	g0011	g0111	g1111
p01			**9, 14, 16, 17**			
p10			**13** (>90)[b]	**11** (>90)[b]		
			19 (20)[c](>20)[d](>20)[e]	**18**		
			24 (140)[c]	**20** (240)[c](>20)[d](17.0)[e]		
			30 (12.2)[d](>20)[e]	**21**		
			31 (>20)[d](>20)[e]	**26** (430)[c](>20)[d](>20)[e]		
			32 (20)[d](>20)[e]	**27** (70)[c](>20)[d](15.5)[e]		
			33 (15.2)[e]	**28** (15.6)[d](>20)[e]		
			35 (>20)[d](>20)[e]	**29** (>20)[d](>20)[e]		
			36	**34** (>20)[d](>20)[e]		
			37 (>20)[d](>20)[e]	**39, 43, 44, 45, 46, 50, 51, 53**		
			38 (>20)[e]	**57** (>20)[d](17.8)[e]		
			40, 55, 56, 65, 68, 70	**59, 60, 61, 62, 63, 64, 66, 69, 71**		
p11	**6** (5.33 ± 0.46)[a](50)[c](8.5)[d] (16.6)[e]	**5** (90)[c](3.4)[d](12.5)[e]	**2** (>30)[b]	**7** (5.02 ± 0.98)[a](>30)[b]	**1** (3.71 ± 0.79)[a]	**3** (2.30 ± 0.38)[a]
		22 (70)[c]	**4** (5.28 ± 0.76)[a]	**15**		
		23 (4.2)[d](2.3)[e]	**8**	**25** (>500)[c](4.4)[d](6.0)[e]		
			10 (>30)[b]	**41, 47, 54**		
			12 (>30)[b]			
			42, 48, 49, 52, 58, 67			

[a] Thang et al., 2013.
[b] Anti-inflammatory activity taken from Ko et al., 2008.
[c] Min et al., 1998.
[d] Min et al., 2000.
[e] Min et al., 2000.

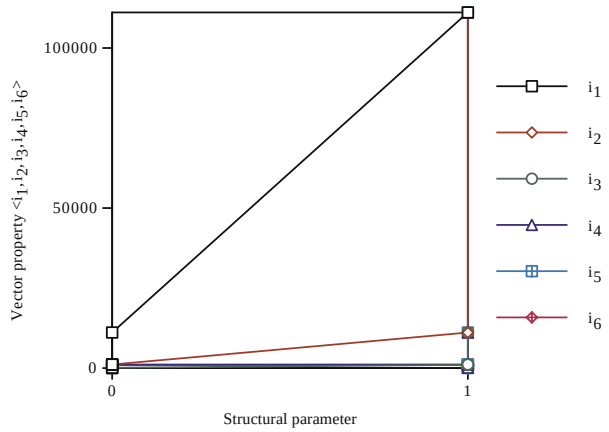


Fig. 4. Variation of the vector property of triterpenoids/steroids in base 10 *vs.* the counts (i_1,i_2,i_3,i_4,i_5,i_6).

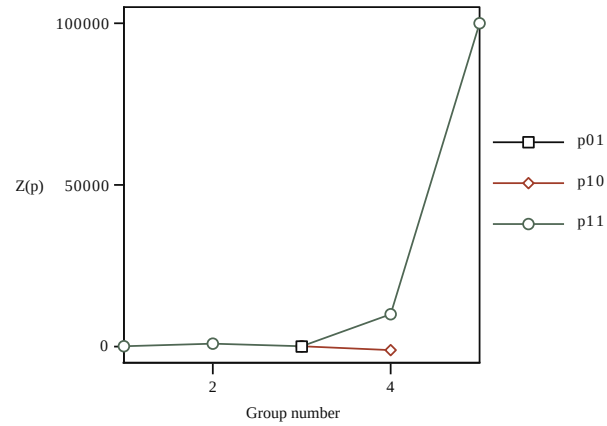


Fig. 6. Variation of $Z(p) = P(p + 1) - P(p)$ *vs.* the group number for triterpenoids and steroids.

Table 3 presents the results of our classification of triterpenoids and steroids in the form of a periodic table (PT). The best result for the PT contains three periods (rows) and six groups (columns) with different values of bioactivities. The compounds are classified first by i_1, then by i_2, i_3 and i_4, all of which represent the group, and, finally, by i_5 and i_6, both of which denote the period. Periods of six units were assumed, which correspond to six groups: *e.g.*, group g0000 stands for <i_1,i_2,i_3,i_4> = < 0 0 0 0>; <0 0 0 0 1 1> (lucidumol B); and group g0001 <0 0 0 1>; <0 0 0 1 1 1> (ganodermanondiol). By definition, several positions in the PT correspond to impossible structures. The molecules in the same column appear close in the partial correlation diagram, dendrogram and PCA (Figs. 1–3). The antioxidant activity increases across the periods (Table 3), and in all cases, it increases when descending in a group. The antioxidant potency also increases through the periods, *e.g.*, in period p11, g0000 < g0001 < g0010 < g0011 < g0111 < g1111. In all periods, it increases when descending in a group, *e.g.*, for g0010 and g0011, p10 < p11.

3.2. Variation of the vector property with the position in the periodic table and assessment of the vector law

Fig. 4 presents the variation of property P of vector <i_1,i_2,i_3,i_4,i_5,i_6>, expressed in the decimal system $P = 10^5 i_1 + 10^4 i_2 + 10^3 i_3 + 10^2 i_4 + 10 i_5 + i_6$, *vs.* the structural parameters {i_1,i_2,i_3,i_4,i_5,i_6} for the triterpenoids and steroids. Some points collapse. For instance, for molecule **1** (Δ^1-lupenone) <0 1 1 1 1 1>, vector property $P = 10^5 \cdot 0 + 10^4 \cdot 1 + 10^3 \cdot 1 + 10^2 \cdot 1 + 10 \cdot 1 + 1 = 11111$, where the structural parameters are 0 and 1 and the corresponding points are ($i_1 = 0$, $P = 11111$) and ($i_2 = i_3 = i_4 = i_5 = i_6 = 1$, $P = 11111$). In accordance with the previously assigned order, parameter i_1 shows greater variation than, for example, i_2 and i_3. The antioxidant activity was not used in the development of the PT (Table 3) and serves to validate it. The results agree with the PT of properties with vertical groups defined by {i_1,i_2,i_3,i_4} and horizontal periods described by {i_5,i_6}.

The variation of property P of vector <i_1,i_2,i_3,i_4,i_5,i_6> in base 10 as a function of the number of the group in the PT for the triterpenoids and steroids (*cf.* Fig. 5) shows that periods p01, p10 and p11 represent rows 1, 2 and 3 in Table 3, respectively. For instance, for group g0010 and period p10, molecule **13** (ganoderic aldehyde A) <0 0 1 0 1 0> lies in the third group with $P = 1010$ at the point (group = 3, $P = 1010$). Some points collapse. Fig. 5 reveals some minima corresponding to compounds with <i_1,i_2,i_3,i_4> close to <0 0 0 0> (g0000 group 1) and maxima near <1 1 1 1> (g1111 group 6). The corresponding function P (i_1,i_2,i_3,i_4,i_5,i_6) reveals a series of periods clearly limited by some minima and maxima, which suggest a periodic *wave* behavior that recalls the form of a trigonometric function. For <i_1,i_2,i_3,i_4,i_5,i_6>, a minimum is clearly shown for most periods. The distance in <i_1,i_2,i_3,i_4,i_5,i_6> units between each pair of consecutive minima is six, which coincides with the number of groups in the PT in successive periods. The minima occupy analogous positions in the corresponding curves, which are in phase. The representative points in phase correspond to the elements of

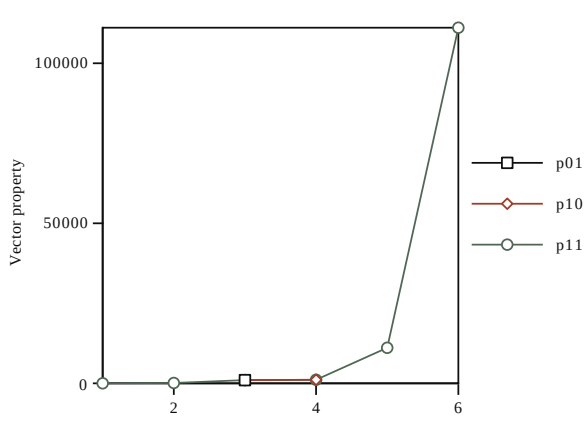


Fig. 5. Variation of the vector property of triterpenoids and steroids in base 10 *vs.* the group number.

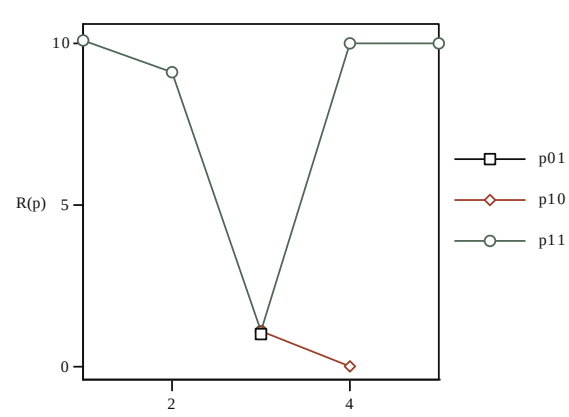
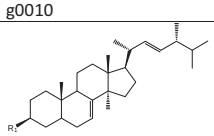
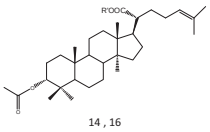
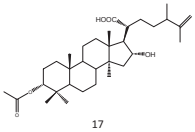
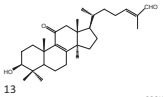
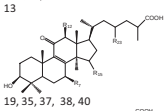
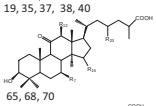
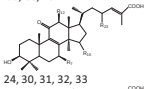
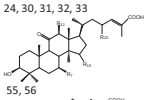
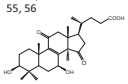
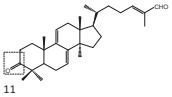
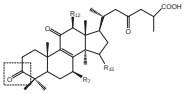
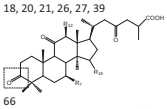
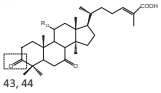
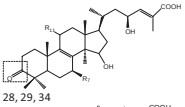
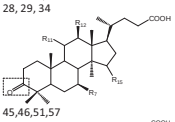
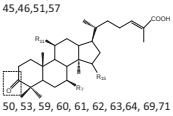


Fig. 7. Variation of $R(p) = P(p + 1)/P(p)$ *vs.* the group number for triterpenoids and steroids.

Table 4
Periodic classification of molecular structures of triterpenoids and steroids from *Ganoderma*.

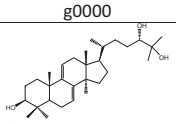
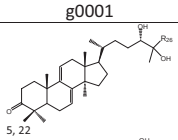
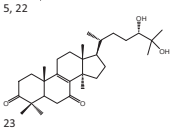
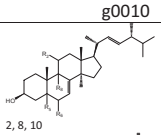
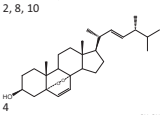
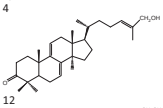
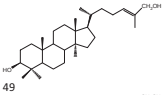
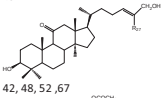
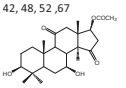
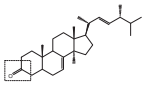
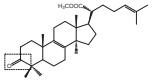
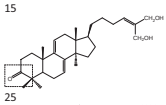
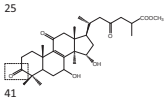
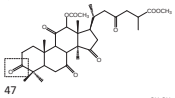
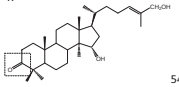
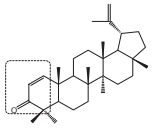
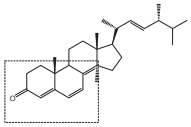
	g0000	g0001	g0010	g0011	g0111	g1111
p01			 9  14 , 16  17			

9 R_1 O-palmitoyl	14 R′: H
	16 R′: β-D-glucosyl

	g0000	g0001	g0010	g0011	g0111	g1111
p10			 13  19, 35, 37, 38, 40  65, 68, 70  24, 30, 31, 32, 33  55, 56  36	 11  18, 20, 21, 26, 27, 39  66  43, 44  28, 29, 34  45,46,51,57  50, 53, 59, 60, 61, 62, 63,64, 69,71		

19 R_7: OH R_{12}: H R_{15}: =O R_{23}: =O 35 R_7: OH R_{12}: H R_{15}: OH R_{23}: =O 37 R_7: =O R_{12}: OH R_{15}: =O R_{23}: =O 38 R_7: OH R_{12}: OH R_{15}: =O R_{23}: =O 40 R_7: =O R_{12}: $OCOCH_3$ R_{15}: =O R_{23}: =O 65 R_7: OH R_{12}: H R_{15}: =O R_{23}: =O	24 R_7: OH R_{12}: H R_{15}: =O R_{23}: 2H 30 R_7: OH R_{12}: H R_{15}: =O R_{23}: OH 31 R_7: =O R_{12}: H R_{15}: =O R_{23}: OH 32 R_7: OH R_{12}:OH R_{15}: =O R_{23}: OH 33 R_7: =O R_{12}: OH R_{15}: =O R_{23}: OH 55 R_7: H R_{12}: H R_{15}: =H R_{23}: 2H	18 R_7: =O R_{12}: =O R_{15}: =O 20 R_7: OH R_{12}: H R_{15}: =O 21 R_7: =O R_{12}: H R_{15}: =O 26 R_7: OH R_{12}: H R_{15}: OH 27 R_7: H R_{12}: H R_{15}: OH 39 R_7: OH R_{12}: H R_{15}: =O	28 R_7:β-OH 29 R_7: α -OH 34 R_7: H 45 R_7: : β- OH R_{11}: =O R_{12}: β- $OCOCH_3$ R_{15}: =O R′:H 46 R_7: H R_{11}: H R_{12}: H R_{15}: α- OH R′:H	50 R_7: =O R_{11}: H R_{15}: H 53 R_7: =O R_{11}: H R_{15}: H 59 R_7: H R_{11}: =O R_{15}: α- OH 60 R_{11}: =O R_{12}: β- $OCOCH_3$ R_{15}: =O 61 R_{11}: =O R_{12}: H R_{15}: α-OH	69 R_7: H R_{11}: H R_{15}: α- $OCOCH_3$ 71 R_7: β- OH R_{11}: =O R_{15}: α- OH

68 R_7: =O R_{12}: H R_{15}: =O R_{23}: =O 70 R_7: OH R_{12}: β-$OCOCH_3$ R_{15}: =O R_{23}:=O	56 R_7: =O R_{12}: H R_{15}: H R_{23}: 2H	66 R_7: H R_{12}: H R_{15}: α–OH 43 R_{11}: α- OH 44 R_{11}: β- OH	51 R_7: =O R_{11}: =O R_{12}: H R_{15}: =O R′:H 57 R_7: β- OH R_{11}: =O R_{12}: H R_{15}: =O R′:H	62 R_7: β- OH R_{11}: =O R_{12}: H R_{15}: α-OH 63 R_7: β- OH R_{11}: =O R_{12}: β- OH R_{15}: =O 64 R_7: =O R_{11}: =O R_{12}: β- OH R_{15}: =O	

	g0000	g0001	g0010	g0011	g0111	g1111
P11	 6	 5, 22  23	 2, 8, 10  4  12  49  42, 48, 52 ,67  58	 7  15  25  41  47  54	 1	 3

5 R_{26}: CH_3 22 R_{26}: CH_2OH	2 R_2: H R_5: H R_6: H R_8: H 8 R_2: H R_5: =O R_6: OH R_8: H 10 R_2: OH R_5: H R_6: H R_8: OH	42 R_7: OH R_4: H R_{15}: =O R_{26}: $COOCH_3$ 48 R_7: OH R_4: $OCOCH_3$ R_{15}: =O R_{26}: $COOCH_3$ 52 R_7: =O R_4: H R_{15}: H R_{26}: $COOCH_3$ 67 R_7: =O R_4: $OCOCH_3$ R_{15}: =O R_{26}: COOH

the same group in the PT. For the <i_1,i_2,i_3,i_4,i_5,i_6>minima, coherence exists between the two representations: the results for all periods present similar trends.

An empirical function $P(p)$ reproduces the different values of <i_1,i_2,i_3,i_4,i_5,i_6>. A minimum of $P(p)$ has meaning only if it is compared with the former $P(p-1)$ and latter $P(p+1)$ points and fulfils the following:

$$\begin{aligned} P_{\min}(p) &< p(p-1) \\ P_{\min} &< (p)(p+1) \end{aligned} \tag{2}$$

Order relationships (2) should repeat at determined intervals equal to the length of the period and are equivalent to

$$\begin{aligned} P_{\min}(p) - P(p-1) &< 0 \\ P(p+1) - P_{\min}(p) &> 0 \end{aligned} \tag{3}$$

Because expressions (3) are valid only for minima, other equations are desired for all values of p. The differences $Z(p)$ are calculated by assigning every value to the triterpenoid or steroid compound at position p in the PT:

$$Z(p) = P(p+1) - P(p) \tag{4}$$

Instead of $Z(p)$, the values of $R(p) = P(p+1)/P(p)$ can be taken by specifying them to the triterpenoid or steroid compound at position p in the PT. If the periodic law (PL) was general, the elements in the same group in analogous positions in different periods (*waves*) would satisfy the following:

$$\text{either } Z(p) > 0 \text{ or } Z(p) < 0 \tag{5}$$

$$\text{and either } R(p) > 1 \text{ or } R(p) < 1 \tag{6}$$

Fig. 6 shows the variation of $Z(p)$ as a function of the number of the group. Some points collapse. For instance, for group g0010 and period p10 in the PT, molecule **13** (ganoderic aldehyde A) <0 0 1 0 1 0> (group = 3, P = 1010) presents, in the next position in the PT, compound **11** (ganoderal A) <0 0 1 1 1 0> (g0011, group = 4, P = 1110), Z = 1110 − 1010 = 100, and the corresponding point is (group = 3, Z = 100). The results are, in general, as expected: $Z(p) > 0$ for groups 1–3 and 5 according to Eq. (5) and the PL. However, the last result in period p10 can be explained because $Z(p)$ for g0011 group 4 is calculated using the first point in the next period.

Moreover, Fig. 7 shows the change of $R(p)$ as a function of the number of the group. Some points collapse. For instance, for group g0010 and period p10, molecule **13** (ganoderic aldehyde A) <0 0 1 0 1 0> (group = 3, P = 1010) shows, in the next PT cell, compound **11** (ganoderal A) <0 0 1 1 1 0> (g0011 group = 4, P = 1110), R = 1110/1010 = 1.099, and the corresponding point is (group = 3, R = 1.099). For most groups (1–3 and 5), $R(p) > 1$ is concordant with Eq. (6) and the PL. Nevertheless, the last result in period p10 can be explained because $R(p)$ is calculated using the first point in the next period.

3.3. Comparison with experiments of antioxidant and other activities from the technical literature

Table 3 shows the antioxidant activity data presented in the technical literature (Ko et al., 2008; Min et al., 1998, 2000; Thang et al., 2013). Because the information was obtained from a great variety of experimental protocols, it is difficult to obtain a unified set of reference data. Our results are in agreement with the bibliography with regard to several types of activities: the HIV-1 protease inhibition of compounds (Min et al., 1998), cytotoxicity of compounds isolated from the spores of *G. lucidum* on tumor cell growth (Min et al., 2000), inhibitory effects of triterpenes and steroids on the formation of $O_2^{\cdot-}$ from rat nucleophiles stimulated with fMLP (Ko et al., 2008), and inhibitory effects of samples from *Ganoderma mastoporum* on the generation of $O_2^{\cdot-}$ by human neutrophils in response to fMLP/CB (Thang et al., 2013). In accordance with the experimental data, the anti-inflammatory activities of triterpenes and steroids from *G. lucidum* and *G. tsugae* (Ko et al., 2008) are predicted to increase for the greater periods (p11), particularly close to the central groups (g0010).

The periodic classification of the molecular structures of triterpenoids and steroids from *Ganoderma*, *cf.* Table 4, indicates that different cells include dissimilar functional groups.

4. Conclusions

This report presents the results of the classification of the 71 studied triterpenoid and steroid compounds from *Ganoderma*. The structural requirements for efficient antioxidant activity received priority. The information entropy and principal component analysis allowed the classification of the substances, and the results agreed. In general, the groups in the periodic table contain constituents with a similar structure and follow the same trend. The antioxidant potency increases through the periods, and in all cases, it also increases when descending in a group.

The groups (columns) on the right side of the periodic table (Tables 3 and 4) are those with the greatest antioxidant activity (*viz.*, g0011, g0111, g1111). The presence of a 3-ketone and the absence of hydroxyl at both positions 24 and 25 are in agreement with the technical literature, which concludes that the 3-ketone is crucial for cytotoxic potency. In contrast, the increase in the number of hydroxyls reduces this activity. Group g1111 with the greatest values of antioxidant potency corresponds to steroids with a 3-ketone conjugated with at least two carbon–carbon double bonds, *e.g.*, ergosta-4,6,8(14),22-tetraen-3-one (**3**). This result can be explained because the presence of these conjugated double bonds stabilizes, by resonance, the corresponding radical and lends planarity to the structure, which increases the activity of these compounds, as was shown in previous publications of flavonoids and stilbenoids (Castellano et al., 2013, 2014).

The triterpenoids and steroids that belong to the same group (Tables 3 and 4, *e.g.*, g0010, g0011) show no apparent difference in bioactivity. Note in period p01 that the 11-esterified triterpenoids or those with a 3-alcoxy group show the lowest antioxidant potency, as confirmed by the experimental data in the bibliography. This finding is due to the following: the corresponding radicals that they generate are possibly destabilized due to a positive mesomeric effect because the alkoxy is a group that, as a result of resonance, is an electron donor.

Furthermore, the presence of either an aldehyde or carboxylic acid at position C-25, despite the fact that these groups present a moderate negative mesomeric effect that could stabilize by resonance the corresponding radical, is not an important factor in the antioxidant activity of the studied compounds, which turn out to be in the central period p10 in the periodic table and thereby present middling antioxidant character.

The antioxidant activity of phenolic compounds, *e.g.*, flavones and coumarins, is universally recognized. However, for *Ganoderma* triterpenoids, their major activities are a result of their anti-inflammatory and anticancer potencies. Hence, antioxidant activity does not appear appropriate for this type of compounds, and the correlation between structure classification and antioxidant potency exhibits a lack of rationale. Therefore, the structure–anti-inflammatory activity relationship is discussed. The anti-inflammatory potency of triterpenes and steroids from *Ganoderma* appears limited to the middle bottom of the periodic table and increase for the greater periods. This activity is predicted to increase for the greater periods (p11) and to be closer to central

groups (g0010). Suitable experimental models should validate the structure–anti-inflammatory activity relationships.

In summary, a set of 71 triterpenoids and steroids from *Ganoderma* was classified with regard to their inhibitory effects on, for instance, the generation of the superoxide anion. Their structure–activity relationships were discussed. As a result, the analysis includes such chemical characteristics of the triterpene and steroid compounds that fit the following general scheme in the hierarchical order: (1) the presence of a 3-ketone conjugated with at least two carbon–carbon double bonds, which were stabilized by resonance favouring a more planar geometry, (2) the presence of a 3-ketone conjugated with at least one carbon–carbon double bond, (3) the absence of a hydroxyl group at both positions 24 and 25, which produced a positive mesomeric effect destabilizing the radical, (4) the presence of a 3-ketone, (5) the presence of either a 3-hydroxyl or 3-ketone, and (6) the absence of both an aldehyde and carboxylic acid at positions 23 and 25.

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